Midterm Exam CS 401/MCS 401 Computer Algorithms I Monday 13 July 2026

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

5. Consider the sequence $S = 10, 56, 91, 42, 21, 68, 46, 67$.

   (a) Compute the number of inversions in this sequence via a divide-and-conquer algorithm. Show all steps.

   (b) In in the worst case, a sequence of n elements may have as many as $O(n^2)$ inversions. Explain why divide-and-conquer leads to a subquadratic cost.

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Answer:

5. (a) Abbreviating `Sort-and-Count` by `S`:

$$
\begin{aligned}
&\texttt{S}(10,\ 56,\ 91,\ 42,\ 21,\ 68,\ 46,\ 67)\\
&= \texttt{S}(10,\ 56,\ 91,\ 42) + \texttt{S}(21,\ 68,\ 46,\ 67)\\
&= \texttt{S}(10,\ 56) + \texttt{S}(91,\ 42) + \texttt{S}(21,\ 68) + \texttt{S}(46,\ 67)\\
&= (0,\ (10,\ 56)) + (1,\ (42,\ 91)) + (0,\ (21,\ 68)) + (0,\ (46,\ 67))\\
&= (1 + 1,\ (10,\ 42,\ 56,\ 91)) + (2,\ (21,\ 46,\ 67,\ 68))\\
&= (2 + 2 + 3 + 2 + 1 + 1,\ (10,\ 21,\ 42,\ 46,\ 56,\ 67,\ 68,\ 91))\\
&= (11,\ (10,\ 21,\ 42,\ 46,\ 56,\ 67,\ 68,\ 91))
\end{aligned}
$$

   (b) The recurrence of the `Sort-and-Count` is that of merge sort: $T(n) = 2T(n/2) + O(n)$.
   The counting of the inversions does not add to the $O(n)$ because we can have at most as many additions as the number of comparisons in the merge stage.
   Therefore, we obtain a subquadratic cost.